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 Project Proposal · Ref: FVT/CBT/2026/001

SUBMITTED TO

Babcock University High School

A proposal for the design, development, and deployment of a secure Computer-Based Testing (CBT) Platform to serve over 2,000 students across all academic levels.

DATE SUBMITTED

May 20, 2026

PREPARED BY

Firmvision Technologies Ltd.

VALID FOR

60 Days from Submission

Confidential — Prepared exclusively for Babcock University High School

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Executive Summary

Firmvision Technologies Limited is pleased to present this proposal for the design, development, and full deployment of a **secure, enterprise-grade Computer-Based Testing (CBT) Platform** tailored exclusively for Babcock University High School.

As one of Nigeria's leading faith-based academic institutions, Babcock University High School maintains rigorous academic standards across a student population of approximately **1,500 learners** — a number expected to grow beyond 2,000 in the coming academic years. Managing internal assessments, continuous assessment tests (CATs), terminal examinations, and promotional examinations at this scale — while preserving integrity, reducing administrative burden, and generating timely results — requires a purpose-built digital solution.

Our proposed **Babcock CBT Platform** delivers a fully-featured examination environment with an administrative control panel for teachers and management, a secure student examination portal, a centralised question bank, automated marking, and a robust results management system. The platform is engineered to operate on the school's existing local area network (LAN) without any dependency on internet connectivity — and can equally be accessed over the internet when required, giving the institution the flexibility of a **hybrid deployment model**.



Our Core Promise

A tamper-proof, efficient, and cost-effective CBT system that eliminates paper-based exam challenges, reduces result-processing time from days to minutes, and gives Babcock University High School full ownership and control of its academic assessment data.

About Firmvision Technologies

Firmvision Technologies Limited is a Nigerian-registered software development and technology consulting company dedicated to building bespoke digital solutions for the education, healthcare, and enterprise sectors. We combine deep technical expertise with a thorough understanding of the Nigerian educational landscape to deliver solutions that are practical, reliable, and built to last.

Our Core Capabilities



Custom Software Development

Web, mobile, and desktop applications built from the ground up to exact specifications.



EdTech Solutions

Learning management systems, CBT platforms, student portals, and academic ERP systems.



Cybersecurity & Data

Secure authentication, encrypted data storage, role-based access control, and audit trails.



Cloud & On-premise Deployment

Solutions that work on local school networks, cloud servers, or both simultaneously.



Analytics & Reporting

Real-time dashboards, result analytics, and exportable reports for school management.



Training & Support

On-site staff training, documentation, and ongoing technical support packages.

Understanding the Challenge

Large secondary schools managing thousands of students face compounding challenges when conducting examinations through traditional paper-based methods. We have identified the following pain points common to institutions of Babcock University High School's scale:

Current Challenges

CHALLENGE AREA	IMPACT
Paper & printing costs	Significant recurring expenditure on question papers, answer sheets, and logistics per examination cycle
Examination malpractice	Question leakage, answer copying, impersonation — difficult to control at scale with paper-based systems
Result processing time	Manual marking and result compilation can take days to weeks, delaying academic progression decisions
Question bank management	No centralised repository — teachers recreate questions repeatedly; no reuse or quality control
Scalability	Simultaneously examining 1,500+ students in a controlled environment is a major logistical challenge
Connectivity dependency	Most cloud-based CBT platforms require constant, stable internet — unreliable in many Nigerian school settings
Result integrity	Paper results are susceptible to tampering, loss, and transcription errors

Proposed Solution Overview

Firmvision Technologies proposes the development and deployment of the **Babcock CBT Platform** — a purpose-built, offline-capable, internet-ready Computer-Based Testing system designed specifically for Babcock University High School.

The platform consists of three integrated components working seamlessly together:



Student Examination Portal

A secure, locked-down examination interface for students. Features fullscreen enforcement, tab-switch detection, countdown timer, auto-save, and automated submission. Designed to work on any modern browser — no app installation required on student devices.



Admin & Teacher Control Panel

A comprehensive management dashboard for school administrators and teachers. Manage students in bulk, build and organise the question bank, create and schedule examinations, monitor live sessions, view detailed results, and manually grade essay questions.



Secure Backend API

A high-performance Python FastAPI backend with role-based authentication, encrypted data storage, question randomisation engine, automatic grading, violation logging, and support for both SQLite (offline/LAN) and PostgreSQL (cloud/enterprise) databases.



Zero-Dependency Design

Unlike platforms requiring constant internet access, all platform components — including the admin panel and student portal — function entirely from the school's local server. No third-party CDN, cloud service, or internet connection is required during examination sessions.

Key Platform Features

Question Bank Management

A centralised, subject-organised repository of examination questions that teachers build once and reuse across multiple examinations.

- Four question types: **Multiple Choice (MCQ)**, **True / False**, **Multi-select**, and **Essay / Short Answer**
- Questions tagged by subject, class level, and topic
- Configurable point values per question
- Optional explanations for each question — displayed after grading
- Full-text search and filter by subject, type, or keyword

Examination Management

- Create exams from question bank with question selection and ordering controls
- Set time limits, start/end windows, and target class groups
- Configurable pass mark percentage per exam
- Questions and answer options shuffled uniquely per student
- **Pre-exam instruction screen** — students are shown a six-step briefing (identity verified, navigation guide, timer rules, fullscreen policy, submission procedure, result display) before the exam begins, ensuring no student can claim ignorance of the rules
- Exam status workflow: Draft → Active → Ended

Student Management

- Register students individually or bulk-import up to 2,000+ via CSV upload
- Students organised by class, registration number, and academic level
- Role-based access: Admin, Teacher, Student
- Account enable/disable without data deletion
- Secure password management with bcrypt encryption
- **Multi-device support** — the student examination portal is fully optimised for desktop computers, laptops, tablets, and smartphones. Mobile users receive a dedicated bottom navigation bar (Previous / Question Counter / Next / Submit) with touch-friendly answer buttons, ensuring examinations can be conducted even in environments where desktop computers are limited

Results & Analytics

- Instant automatic grading for MCQ, True/False, and Multi-select questions
- Manual essay grading interface with partial marks support
- Per-student result cards showing score, percentage, time taken, and violation log
- Class-wide results table sortable by score, name, or class
- Average score, pass rate, and highest/lowest scores per examination
- **Score distribution analytics chart** — a colour-coded bar chart (red below 50%, amber 50–69%, green 70%+) showing how the class performed across ten percentage bands, giving teachers an instant visual overview of examination difficulty and class preparedness
- **Printable student result slip** — immediately after submission, students can print a formal result slip showing their name, examination title, score, percentage, pass/fail status, date and time, and signature lines for school stamp and countersignature — eliminating the need to manually transcribe results onto paper
- Export-ready data for school record systems

Security & Examination Integrity

Examination integrity is the cornerstone of any credible assessment system. The Babcock CBT Platform employs a multi-layered security architecture at both the technical and procedural level.



Fullscreen Enforcement

The exam automatically requests fullscreen mode on start. Exiting fullscreen logs a violation and triggers a warning. Repeated exits auto-submit the exam.



Tab-Switch Detection

Any navigation away from the exam tab or window is instantly detected, logged, and reported. Students are warned; repeated switches trigger auto-submission.



Question & Option Randomisation

Each student receives questions and answer options in a unique random order, making sharing answers between adjacent students impossible.



One-Time Session Tokens

A unique cryptographic token is issued per student per exam. The same student cannot open the same exam on two different devices simultaneously.



Copy & Right-Click Disabled

During active examinations, clipboard access, right-click menus, and browser developer tools are blocked to prevent question extraction.



Server-Authoritative Timer

Exam time is tracked server-side and verified via regular heartbeat calls. Manipulating the browser clock has no effect on the exam timer.



JWT Authentication & Bcrypt

All sessions use industry-standard JSON Web Tokens. Passwords are stored using bcrypt hashing — never in plain text. All API endpoints are protected.



Violation Audit Log

Every security event is timestamped and stored against the student's session. Administrators can review the full violation history for any submission.

Deployment Options — Offline & Online

A key differentiator of the Babcock CBT Platform is its ability to operate in **both offline (LAN) and online (internet) environments** without any code changes. The school can choose either mode or run them simultaneously.

MODE 1 — RECOMMENDED

School LAN (Offline)

The platform server runs on a single school computer or dedicated server. All students connect through the school's Wi-Fi or LAN cable. **No internet access required at any point.**

- ✓ Works on existing school Wi-Fi infrastructure
- ✓ All data stays within the school premises
- ✓ SQLite database — zero additional software
- ✓ Supports 500+ concurrent users on a standard PC
- ✓ Single command to start:
`./start.sh`
- ✓ Ideal for terminal exams and internal assessments

MODE 2

Cloud / Internet

Deployed on a cloud server (e.g. a VPS or dedicated server), the platform is accessible from anywhere — school, home, or mobile — over the internet.

- ✓ Accessible from any device, anywhere
- ✓ PostgreSQL database for high-volume concurrency
- ✓ Docker deployment for easy management
- ✓ Encrypted HTTPS for all data in transit
- ✓ Suitable for remote/take-home assessments
- ✓ Daily automated database backups



Hybrid Recommendation

For maximum reliability, we recommend the LAN-primary, internet-backup model: conduct all formal examinations on the school's local network (zero dependency on internet provider), while keeping the cloud deployment available for remote access by school management and reporting.

Technical Architecture

The platform is built on a modern, proven technology stack selected for performance, security, long-term maintainability, and independence from expensive proprietary software licenses.

LAYER	TECHNOLOGY	WHY
Backend API	Python 3.12 + FastAPI	Async I/O handles thousands of concurrent exam sessions; auto-generates API documentation; type-safe
Database (LAN)	SQLite	Zero installation; single file; works offline; easily backed up; sufficient for 2,000+ students
Database (Cloud)	PostgreSQL 16	Enterprise-grade; handles 10,000+ concurrent connections; full ACID compliance; free, open-source
Authentication	JWT + bcrypt	Industry standard; stateless tokens; passwords never stored in plain text
Admin Frontend	Vanilla HTML/ CSS/JavaScript	Zero CDN or internet dependency; loads instantly on any device; no framework licensing costs
Student Frontend	Vanilla HTML/ CSS/JavaScript	Lightweight, reliable under low-bandwidth LAN conditions; works on any browser including older devices
Containerisation	Docker + Docker Compose	Reproducible, portable deployment; the entire system runs with a single command

LAYER	TECHNOLOGY	WHY
Web Server	Uvicorn (ASGI)	High-performance Python server; handles thousands of requests per second

Minimum Server Requirements (LAN Mode)

COMPONENT	MINIMUM	RECOMMENDED
Processor	Intel Core i3 / equivalent	Intel Core i5 / i7 or better
RAM	4 GB	8 GB or more
Storage	50 GB HDD	120 GB SSD
Operating System	Windows 10, Ubuntu 20.04	Ubuntu 22.04 LTS (Linux)
Network	100 Mbps switch	Gigabit switch + structured cabling
Python	Python 3.10	Python 3.12

Implementation Timeline

From contract signing to full go-live, the project is structured into four phases over an estimated **6–8 weeks**.

Weeks 1 – 2

PHASE 1 — DISCOVERY & SETUP

Requirements Finalisation & Infrastructure Setup

Kick-off meeting with school management and IT team. Finalise subject list, class structure, user roles, and examination policies. Assess existing network infrastructure. Set up the server environment (school-provided hardware or cloud, per agreement). Install and configure all platform components. Create initial admin accounts.

Weeks 3 – 4

PHASE 2 — DATA MIGRATION & CONFIGURATION

Student Registration & Question Bank Loading

Bulk import of all student records (name, class, registration number) from school register. Subject configuration for all classes and academic levels. Teachers receive accounts and begin loading questions into the bank with Firmvision technical support on-site. Platform customised with school branding (name, logo, colour scheme).

Weeks 5 – 6

PHASE 3 — TRAINING & PILOT TESTING

Staff Training & Mock Examination

Structured training for administrators, teachers, and IT staff on all platform functions. Printed user guide and quick-reference cards for teachers. Conduct a supervised pilot examination with a sample student group (one class or year group) to validate the full workflow from question loading to result generation. Feedback incorporated and issues resolved.

Weeks 7 – 8

PHASE 4 — GO-LIVE & HANDOVER

Full Deployment & Official Launch

Full school-wide examination conducted on the platform. Firmvision technical team present on-site for the first official examination cycle. Formal handover documentation, admin credentials, backup procedures, and disaster recovery guide delivered. 90-day intensive post-launch support period commences.

Investment & Pricing

All packages include the complete platform as described in this proposal. The difference between packages lies in support depth, additional features, and maintenance coverage. All prices are in Nigerian Naira (₦) and exclude VAT.

Standard ₦ 1,800,000 One-time setup fee Ideal for getting started — LAN deployment with essential features. ✓ Full CBT platform (LAN only) ✓ Up to 3,000 student accounts ✓ Question bank & 4 question types ✓ Automated result & grading ✓ 3-day on-site staff training ✓ 30-day post-launch support ✓ 1-year software updates + ₦400,000/year maintenance	RECOMMENDED Professional ₦ 2,800,000 One-time setup fee Full-featured LAN + cloud hybrid deployment for up to 6,000 students. ✓ LAN + cloud hybrid deployment ✓ Up to 6,000 student accounts ✓ All Standard features ✓ PostgreSQL (cloud) database ✓ HTTPS/SSL certificate setup ✓ 5-day on-site staff training ✓ 90-day post-launch support ✓ 2-year software updates ✓ Results analytics dashboard ✓ Priority support (4-hr response)	Enterprise ₦ 4,500,000 One-time setup fee Complete solution for the entire school with full customisation and dedicated SLA. ✓ All Professional features ✓ Unlimited student accounts ✓ Custom school branding & domain ✓ Parent result portal ✓ Dedicated server setup & config ✓ Full network audit & hardening ✓ 7-day training programme ✓ 1-year post-launch support ✓ 3-year software updates ✓ 2-hour emergency SLA + ₦900,000/year maintenance
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+ ₦600,000/year
maintenance

Payment Terms

MILESTONE	PAYMENT	WHEN
Mobilisation	50% of agreed package fee	Upon contract signing
Pilot completion	30% of agreed package fee	After successful pilot examination
Go-live	20% of agreed package fee	After official launch and handover



Flexible Financing Available

We understand budget planning cycles in educational institutions. Firmvision is open to discussing structured payment plans that align with the school's term-based budget releases, subject to mutual agreement.

Support & Maintenance

A technology platform is only as valuable as the support behind it. Firmvision provides comprehensive post-deployment support to ensure continuous, uninterrupted operation.



Helpdesk Support

Dedicated phone and email support line for teachers and IT staff during school hours (7 AM – 6 PM)



Software Updates

Regular platform updates including bug fixes, security patches, and new feature releases



Data Backups

Automated daily database backups with tested recovery procedures. Backup copies stored securely



Refresher Training

Annual one-day refresher session for new staff and teachers joining after initial deployment



Exam-Day Support

Remote (and on-site for Enterprise) technical standby during scheduled major examination periods



Quarterly Reviews

Quarterly check-in meetings to review system performance, address feedback, and plan enhancements

Why Firmvision Technologies

1 Built for Nigerian Schools

We understand the realities of Nigerian school infrastructure — unreliable internet, mixed device types, and varying technical literacy. Our solution is engineered for these conditions, not imported from foreign contexts.

2 Full Source Code Ownership

Unlike SaaS subscription platforms that hold your data hostage, Babcock University High School retains full ownership of the platform and all student data. There are no recurring licence fees or vendor lock-in risks.

3 Purpose-Built, Not Generic

This is not an off-the-shelf product with unwanted features. Every element of the Babcock CBT Platform is built specifically for your school's assessment workflows, class structures, and examination calendar.

4 Proven Security Architecture

Multi-layer examination integrity controls — fullscreen enforcement, tab detection, one-time tokens, question randomisation — backed by cryptographic authentication and a complete violation audit trail.

5 Zero Recurring Licence Costs

The platform is built on open-source technology (Python, FastAPI, PostgreSQL). After the one-time setup fee, there are no per-user, per-exam, or per-month licensing charges — ever.

6 Long-Term Partnership

We do not disappear after go-live. Our support packages, quarterly reviews, and refresher training programmes are designed to ensure the school's IT team grows in confidence with the platform over time.

Terms & Conditions

1. This proposal is valid for **60 calendar days** from the date of submission (i.e., until 19 July 2026). After this period, pricing is subject to review.
2. All prices are exclusive of Value Added Tax (VAT) at the prevailing rate. VAT will be itemised separately on the invoice.
3. The scope of work is as described in this proposal. Any additions or modifications requested after contract signing will be subject to a formal change order and additional charges.
4. The school undertakes to provide access to necessary infrastructure, including server hardware (or cloud subscription), school network access, and student registration data in the agreed format.
5. Firmvision Technologies will treat all student data, examination content, and institutional information with strict confidentiality and in compliance with applicable Nigerian data protection regulations (NDPR).
6. Intellectual property of the platform codebase will transfer fully to Babcock University High School upon receipt of the final payment milestone.
7. Annual maintenance fees cover the services described in Section 11. Failure to renew the maintenance agreement does not affect the operation of the installed platform but will suspend support services.
8. Firmvision Technologies shall not be held liable for examination integrity failures arising from school-side procedural breaches (e.g., students sharing login credentials) or failures of school-provided hardware and network infrastructure.
9. Either party may terminate the agreement before the pilot phase with 14 days written notice. In such cases, the mobilisation payment (50%) is non-refundable to cover work already commenced.
10. This proposal supersedes all prior verbal or written discussions between Firmvision Technologies and Babcock University High School regarding this project.

Acceptance & Signatures

By signing below, both parties confirm their understanding and acceptance of the terms set out in this proposal and agree to proceed with contract formalisation.

For Firmvision Technologies Limited

Proposing Party

Signature: _____

Name: _____

Title: Director / Authorised Signatory

Date: _____

Stamp:

For Babcock University High School

Receiving Party

Signature: _____

Name: _____

Title: Principal / Authorised Officer

Date: _____

Stamp:

For enquiries regarding this proposal, please contact:

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